

blockchain provides a trustworthy foundation for AI by ensuring the transparency and auditability of AI operations through its immutable record-keeping. It also offers a secure environment for managing the data to train AI models and enables privacy-preserving AI computations.

AI makes blockchain smarter

Supercharged security: AI can act like a digital detective, spotting unusual activity on blockchain networks that might signal fraud or attacks. It can also help find weaknesses in the blockchain application code, making them safer.

Faster and more efficient: One of blockchain's challenges has been handling many transactions quickly. AI can step in to optimise how these transactions are processed, making things faster and using less energy.

Smarter agreements: Imagine contracts that can think for themselves! AI is helping to create 'smart contracts' that can analyse data, predict outcomes, and make decisions automatically, going way beyond the simple rules of traditional contracts.

Data detective: Blockchain holds a treasure trove of data but can be tricky to analyse. AI can act as a data detective, helping to find patterns and insights hidden within all that information.

Blockchain makes AI more trustworthy

See-through AI: One of the worries about AI is that it can be like a 'black box' – we don't always know how it makes decisions. Blockchain can provide a transparent record of AI's actions, showing exactly what data it used and how it reached its conclusions. This builds trust in AI systems.

Secure data for AI: AI needs reliable data to learn from. Blockchain can provide a secure and unchangeable place to store this data, ensuring it hasn't been tampered with. This is crucial for AI to make accurate predictions.

Keeping data private: Sometimes, the data AI uses is sensitive, like medical records. Blockchain and special privacy techniques can allow AI to work with this data without revealing private information.

AI for everyone: Blockchain can help create open platforms where AI models, data, and computing power can be shared securely and transparently. This could make AI more accessible to everyone.

Blockchain has the potential to foster the development of decentralised AI platforms and marketplaces, making AI more accessible. This mutually beneficial relationship drives innovation and creates new possibilities across various industries.

Key characteristics of blockchain and AI

Blockchain technology's fundamental features include its decentralised nature, where data is distributed across a network; immutability, ensuring records cannot be altered; and transparency, allowing participants to view transactions. It operates as a distributed ledger, employing cryptographic security and consensus mechanisms to validate transactions and maintain data integrity. Smart contracts and self-executing agreements are key applications. These features enable blockchain to be applied across various industries, including cryptocurrencies, decentralised finance

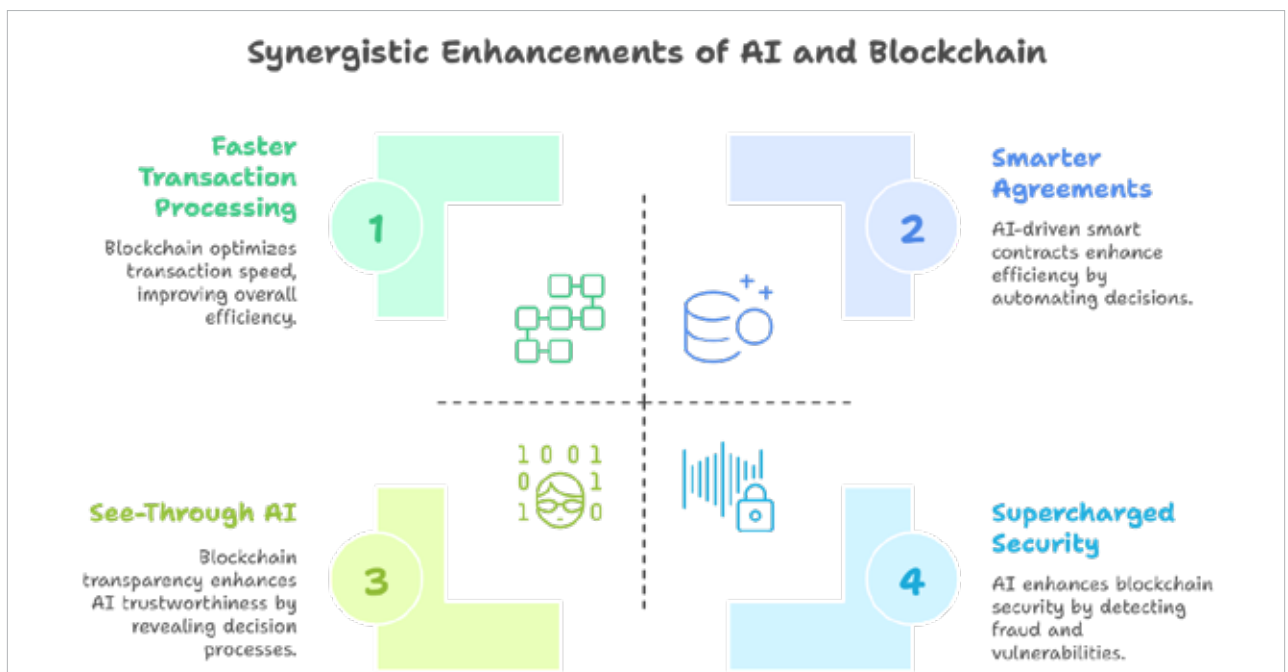


Figure 1: Dynamic synergistic duo between blockchain and AI